

STATISTICS FOR MANAGEMENT.

UNIT I INTRODUCTION

Basic definitions and rules for probability, conditional probability independence of events, Baye's theorem, and random variables, Probability distributions: Binomial, Poisson, Uniform and Normal distributions.

UNIT II SAMPLING DISTRIBUTION AND ESTIMATION

Introduction to sampling distributions, sampling distribution of mean and proportion, application of central limit theorem, sampling techniques. Estimation: Point and Interval estimates for population parameters of large sample and small samples, determining the sample size.

UNIT III TESTING OF HYPOTHESIS - PARAMETRIC TESTS Hypothesis testing:

one sample and two sample tests for means and proportions of large samples (ztest), one sample and two sample tests for means of small samples (t-test), F-test for two sample standard deviations. ANOVA one and two way

UNIT IV NON-PARAMETRIC TESTS

Chi-square test for single sample standard deviation. Chi-square tests for independence of attributes and goodness of fit. Sign test for paired data. Rank sum test. Kolmogorov-Smirnov – test for goodness of fit, comparing two populations. Mann – Whitney U test and Kruskal Wallis test. One sample run test.

UNIT V CORRELATION AND REGRESSION

Correlation – Coefficient of Determination – Rank Correlation – Regression – Estimation of Regression line – Method of Least Squares – Standard Error of estimate.

Unit I – INTRODUCTION

Binomial Distribution

- 1) I) The Mean & S.D of binomial distribution are 5 and 2. Determine the distribution.
ii) Find the binomial distribution for which the mean is 4 and variance is 3.
Link : https://youtu.be/sIIT_GcQfjM?si=WPY9K7-bJNC-3KGS
- 2) Check Binomial distribution or not?
Check whether the following data follow a binomial distribution or not Mean =3, variance =4,
Link : <https://youtu.be/4RJ0SIB5Mfc?si=YP14XysRNtmyYJWV>
- 3) Find Mean and Variance of Binomial Distribution.
Link : <https://youtu.be/ZrRjSd4IOEM?si=i5a3RdW4rtHOsJ9L>
- 4) Find the Moment Generating Function (MGF) of a binomial distribution.
Link : <https://youtu.be/PsoVFYW3jCg?si=3Nezm7jGpDxUUaIC>
- 5) Binomial Distribution Problems:
 - i) 10 coins are thrown simultaneously. Find the probability of getting atleast 7 heads.
 - ii) 6 dice are thrown 729 times. Howmany times do you expect atleast three dice to show a five or six?
Link: https://youtu.be/j5O4u2nHdLA?si=q1Ijlx_aYgnkDKLM
- 6) 4 coins were tossed simultaneously. What is the probability that of getting i) 2 heads, ii) Atleast 2 heads, iii) atmost 2 heads.
Link : https://youtu.be/fo4IKWjWhSk?si=gd_T71zmK0_7ftps
- 7) In a large consignment of electric bulbs 10% are defective. A random sample of 20 is taken for inspection. Find the probability that, i) All are good bulbs, ii) Atmost there are 3 defective bulbs, iii) Exactly there are 3 defective bulbs.
Link : <https://youtu.be/AJeIs2HqvLo?si=WzmtFQrFm2rXEnLn>
- 8) I) A perfect cubic die is thrown a large number of times is sets of 8. The presence of a 5 or 6 is treated as a success. In what percentage of the sets can we expect 3 success.
ii) Five fair coins are flipped. If the outcomes are assumed independent find the probability mass function of the number of heads obtained.
iii) The mean and variance of binomial distribution are 8 & 6. Find $P(X \geq 2)$.
Link : <https://youtu.be/5BLRAWr8ZhQ?si=LbpmFCJYvKKdnJqh>

Poisson Distribution

1) Find Mean and variance of Poisson distribution:

Link: <https://youtu.be/k4d1inrgwCA?si=WwmXq3GdNbkJ26e>

2) Find MGF of Poisson Distribution:

Link : <https://youtu.be/tmuqt4MUWrM?si=9lom2bjrdYMoxjw4>

3) I) Write down the probability mass function of the Poisson distribution which is approximately equivalent to $B(100, 0.02)$.

ii) If 3% of the electric bulbs manufactured by a company are defective, find the probability that in a sample of 100 bulbs exactly 5 bulbs are defective ($e^{-3} = 0.0498$).

iii) If it is known from the past experience that in a certain plant there are on the average 4 industrial accidents. Find the probability that in a given year there will be less than 4 accidents.

Link : https://youtu.be/bxswMphJJF4?si=QZAgsg_qQmV5v87o

4) i) Assume that the chance of an individual coal miner being killed in an accident during a year is $1/1400$. Using a Poisson distribution calculate the probability that in a mine employing 350 miners, there will be atleast one fatal accident in a year. ($e^{-0.25} = 0.7788$).

ii) Find the probability that atmost 5 defective fuses will be found in a box of 200 fuses. If experiences shows that 2% of such fuses are defective.

iii) It is known that probability that an item produced by a certain machine, will be defective is 0.01. By applying Poisson's distribution, show that the probability that a random sample of 100 items selected at random from total output will contain not more than one defective items is $2/e$.

Link : https://youtu.be/m42NPtLzySE?si=JtG-PmKs8-9_SdVM

5) i) A Manufacturer of pins knows that 2% of his products are defective. If he sells pins in boxes of 100 & gurantees that not more than 4 pins will be defective what is the probability that a box will fail to meet the guranted f quality?

ii) If it is known from the past experience that in a certain plant there are on the average 4 industrial accidents. Find the probability that in a given year there will be less than 4 accidents.

Link: https://youtu.be/hMMS9Ms_UUQ?si=TZciz9qtukkj1u76

6) I) The number of monthly breakdown of a computer is a random variable having a poisson distribution with mean equal to 1.8 Find the probability that this computer will function for a month with only one breakdown.

ii) One percent of jobs arriving at a computer system need to wait until weekends for scheduling owing to core-size limitations. Find the probability that among a sample of 200 jobs there are no jobs that have to wait until weekends.

iii) If 3% of the electric bulbs manufactured by a company are defective, find the probability that in a sample of 100 bulbs exactly 5 bulbs are defective ($e^{-3} = 0.0498$).

iv) In turning our certain toys in a manufacturing process in a factory, the average number of defectives is 10% what is the probability of getting exactly 3 defective toys in a sample of 10 boys chosen at random by using the poisson distribution.

Link: <https://youtu.be/WmIckxYdOSE?si=FPRPK8pxMGDdAuIh>

Geometric & Uniform Distribution

1) Find Mean and variance of Geometric Distribution:

Link: <https://youtu.be/x-K9-QHWuss?si=Hqvv4wudrYU2Xdcp>

2) To find MGF of Geometric Distribution:

Link : https://youtu.be/drNEqzs_zQI?si=TdLD0VtvjMFU_nP6

3) If the probability that an applicant for a driver's licence will pass the road test on any given trial is 0.8, what is the probability that he will finally pass the test a) On the fourth trial b) in fewer than 4 trials?

Link : https://youtu.be/WY3V_dwx2jM

4) Geometric Distribution 2 Mark Questions:

Link: <https://youtu.be/Zuu7rAeRBoU?si=z1YslldCsG5joaeS>

Uniform Distributions

1) Find Mean and variance of Uniform Distribution:

Link: https://youtu.be/B_qd-tta6xY?si=JAtUlQlqO65RAJU

2) Find MGF of Uniform Distribution:

Link : <https://youtu.be/SagZvdvhtH8?si=MQe2GtZcaiyviCk->

3) Let X be uniformly distributed over the interval $(-2,2)$. Find the mean, variance and moment generating function of X.

Link : https://youtu.be/28FfRjMaHSw?si=s3iPXGSpe_9bsYcq

4) If X is uniformly distributed over $(0,10)$ find the probability that $X < 2$, $X > 8$, $3 < X < 9$.

Link : <https://youtu.be/TYLvxm0-phk?si=Ga5tuIim9VDQYF4d>

Exponential, Gamma & Normal Distribution

Exponential Distribution :

1) To find MGF of Exponential Distribution:

Link : <https://youtu.be/U6HUARbOrW4?si=GAjPotSCrp9Ooagq>

2) Find Mean and variance of Exponential Distribution:

Link: <https://youtu.be/wAOTQ5HuKHc?si=UqScWn0ZaznZv4U9>

3) Exponential Distribution- 2 Mark Questions:

Link: https://youtu.be/msm7tzgjJco?si=htE9xcqX8kavo_eK

Gamma Distribution Problems:

- 1) To find MGF of Gamma Distribution:

Link : <https://youtu.be/gSUVJRJnFem8?si=dRxsgbqg16EVT14->

- 2) Find Mean and Variance of Gamma Distribution:

Link : <https://youtu.be/WFQTdERuJcE?si=o4SfPTcXxJXhBvBL>

Normal Distribution Problems:

- 1) To find MGF of Normal Distribution:

Link : <https://youtu.be/30DYcvZ2CDw?si=lwjEatA7ceB4rGaE>

- 2) Find Mean and Variance of Normal distribution:

Link: <https://youtu.be/tP7wNM7dlyo?si=UpETzeJTysSYE8-O>

- 3) Normal Distribution – 2 Mark Questions:

Link: <https://youtu.be/pwL5Lwdk70E?si=Y-g0iyo5Gd2HJvqi>

Probability

- 1) **Axioms of Probability:** - Theorem and Results

Link : <https://youtu.be/l84vEYuMCIs?si=gtbKHGd8ulCz8y6x>

- 2) The weekly wages of 1000 workmen are normally distributed around a mean of Rs. 70 and with standard deviation of Rs. 5. Estimate the number of workers whose weekly wages will be between Rs. 69 and Rs. 72; less than Rs.69; more than Rs 72.

Link : https://youtu.be/tNR0e09gR0U?si=OJUXNI-V-W_whIqr
<https://youtu.be/aXOzlesLMKQ?si=1HNootnm6BybD-T5>

- 3) Find C.D.F Given the pdf of a continuous random variables 'X' as follows;

$$f(x) = \begin{cases} 6x(x-1), & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases} \text{ Find C.D.F for 'X'.$$

Link : <https://youtu.be/-cxI1bOe88g?si=-KkoylULCyz7Zbst>

- 4) Let the random variable X assume the value r with the probability law :

$$P(X = r) = q^{r-1}p; \quad r = 1, 2, 3, \dots \text{ Find the moment generating function of X and hence its mean.}$$

Link : https://youtu.be/3Lue_M4v_wk?si=pg7P4rWKPUJDPeMA

- 5) An insurance company found that only 0.005% of the population is involved in a certain type of accident each year. If its 2000 policyholders were randomly selected from the population. What is the probability that not more than two of its clients are involved in such an accident next year?

Link : https://youtu.be/-zUrie5JPII?si=X_x2JKIYs-eIV7FT

- 6) Let X be a random variable with pdf $f(x) = \frac{1}{3} e^{-x/3}, x \geq 0$. Find the moment generating function of X and hence find its mean and variance.

Link : https://youtu.be/kC-Wtj24qWo?si=9nfgOHIdCglZC_Id

Baye's Theorem:

- 1) The Contents of Urns I, II, III are as follows; 1 white, 2 black & 3 red balls; 2 white, 1 black & 1 red balls, 4 white, 5 black & 3 red balls one urn is chosen at random and two balls drawn from it. They happen to be white and red. What is the probability that they come from Urns I, II and III?

Link: <https://youtu.be/4F5ogr4tgr4?si=ukgBRh572jtKx3BN>

- 2) BAYE's Theorem Problems:

Link:

Unit III – TESTING OF HYPOTHESIS PARAMETRIC TESTS

Normal Distribution

Type 1 – Single Mean value problems:

- 1) A sample of 900 members has a mean 3.4 cm & S.D is 2.61 cm. Is the sample from a large population of mean 3.25cms & S.D of 2.61cms. Test 5% level of significance. The value of z at 5% level is $|z_\alpha| < 1.96$

Link: https://youtu.be/gbTOFZa6_MU?si=5-Um7ebDj2Lpewrv

- 2) The mean lifetime of a sample of 100 light bulbs produced by a company is computed to be 1570 hrs with s.d of 120 hrs. If μ is mean lifetime of all the bulbs produced by the company, test the hypothesis $\mu = 1600$ hrs, against the alternative hypothesis $\mu \neq 1600$ hrs with $\alpha = 0.05$ & 0.01 .

Link : <https://youtu.be/Bd3TyQONgfA?si=VJjCtzszQvFdX1Mn>

Type II – Difference of Mean value problems:

- 3) The mean height of two samples of 1000 and 2000 members are respectively 67.5 & 68 inches. Can they be regarded as drawn from the same population with S.D 2.5 inches.

ii) The following table gives the data on the hardness of wood stored outside and inside room.

Sample Size	Outside	Inside
	40	110
Sample Mean	117	132
Sum of squares of deviation of Mean	8655	27244

Test Significance or not.

- iii) Two samples of sizes 400 and 500 have mean 10.9 and 11.5 respectively. Can the samples be regarded as drawn from same population with variance 25?

Link: <https://youtu.be/5WK9dL6Ivug?si=eUnErhBerkWCkWPR>

t-test Distribution

Type 1 – Single Mean value problems:

1. Given a sample mean of 83, a sample S.D of 12.5 and a sample size of 22, test the hypothesis that the value of the population mean is 70 against the alternative that it is more than 70. Use the 0.025 significance level.

ii) A Machinist is making engine parts with axle diameter of 0.7 inches. A random sample of 10 parts shows a mean diameter of 0.742 inch with a standard deviation of 0.04 inches. Compute the statistic you would use to test whether the work is meeting specification.

Link: <https://youtu.be/O2KDVqW4s1k?si=jPwosJJkC4fgX6n0>

2. A certain injection administered to each of 12 patients resulted in the following increases of blood pressure: 5, 2, 8, -1, 3, 0, 6, -2, 1, 5, 0. Can it be concluded that the injection will be, in general accompanied by an increase in B.P.,?

ii) A random sample of 10 boys had the following I.Q's 70, 120, 110, 101, 88, 83, 95, 98, 107, 100. Do these data support the assumption of a population mean I.Q of 100?

Link: https://youtu.be/hDOSITkm_GA?si=oHDEyS5D6IE9OYY2

3. A Sample of 26 bulbs gives a mean life of 990 hours with SD of 20 hours. The manufacturer claim the mean life of bulbs is 1000 hours. Is the sample not upto the standard.

Link: <https://youtu.be/AARhrtkNdxs>

4. A random sample of size 16 valves from normal distribution showed mean of 53 and sum of squares of deviation from this mean equals to 150. Can this sample be regarded as taken from the population having 56 as mean. Obtain 95% confidence limits of the mean of the population.

Link: <https://youtu.be/IMJUHfcTb9I>

5. The mean weekly sales of soap bars in departmental store was 146.3 bars per store. After an advertising campaign the mean weekly sales in 22 stores for typical week increased to 153.7 showed S.D of 17.2 was the advertising campaign successful.

Link : <https://youtu.be/JEoHGuLcfZs>

6. A Machine is designed to produce insulting washers for electrical devices of average thickness of 0.025 cm. A random sample 10 washers was found to have a thickness of 0.024 cm & S.D of 0.002cm. Test the significant of the deviation. Table of t for a degree of freedom at 5% level is 2.262

Link: https://youtu.be/1z_PG9CItuU

F - Distribution

Type 1: Table value problems (To find variance)

1. A group of 10 rat fed on diet A and another group of 8 rat fed on diet B recorded the following increase in weights:

Diet A	5	6	8	1	12	4	3	9	6	10
Diet B	2	3	6	8	10	1	2	8		

Find if the variances are significantly different.

ii) Two independent samples of sizes 9 & 7 from a normal population had the following values of the variables

Sample 1	18	13	12	15	12	14	16	14	15
Sample 2	16	19	13	16	18	13	15		

Do the estimates of the population variance differ significantly at 5% level?

Link: <https://youtu.be/A8UrR-JZmt8?si=fnlrETWwxJ5HS393>

Type 2: Theory problems (Sum of squares of)

i) In a sample of 8 observations, the sum of the squares of deviations of the sample values, from the sample mean was 84.4 and in the other sample of 10 observations was 102.6 Test whether this difference is significant at 5% level, given that the 5 percent point of F for $v_1 = 7$ and $v_2 = 9$ degree of freedom is 3.29

ii) In one sample of 10 observations, the sum of the squares of the deviations of the sample values from the sample mean was 120 and in another sample of 12 observations it was 314. Test whether this difference is significant at 5% level of significance.

Link : <https://youtu.be/Mp-JACMRkiY?si=tKZzbk8CcYBt3f9X>

1) One-way ANOVA Working Procedure

Link: <https://youtu.be/YAwTRBoeNUI>

Unit IV – TESTING OF HYPOTHESIS NON - PARAMETRIC TESTS

Chi Square Distribution

Type 1: Direct Method

1) A random sample of size 25 from a population gives the sample standard deviation 8.5. Test the hypothesis that the population s.d is 10.

Link : <https://youtu.be/iLUdV9frCTQ?si=hJwQtVy2iuIDbbLI>

2) The S.D of the distribution of time taken by 15 workers for performing a job is 6.4 sec. Can it be taken as a sample from a population whose s.d is 5 sec.

Link : <https://youtu.be/RQpwjxnTc3I?si=5maKQOROCGZZSaCf>

Type 2: Goodness of fit (Coin based on Problems) – Binomial Distribution

i) Five coins are tossed 320 times. The number of heads observed is given below

Number of Heads	0	1	2	3	4	5
Frequency	15	45	85	95	60	20

Examine whether the coin is unbiased. Use 5% level of Significance.

ii) 4 coins were tossed 160 times and the following results were obtained

No.of Heads	0	1	2	3	4
Observed Frequencies	17	52	54	31	6

Under the assumption that the coins are unbiased, find the expected frequencies of getting 0,1,2,3,4 heads and test the goodness of fit.

Link : https://youtu.be/7fN_lwsER8w?si=KpY8cFIB19sk0zNW

Type 3: Goodness of fit (Days based on Problems)

- i) A company keeps records of accidents. During a recent safety review, a random sample of 60 accidents was selected and classified by the day of the week on which they occurred.

Day	Mon	Tue	Wed	Thu	Fri
No.of accidents	8	12	9	14	17

Test whether there is any evidence that accidents are more likely on some days than others.

- ii) The following data gives the number of aircraft accidents that occurred during the various days of a week. Find whether the accidents are uniformly distributed over the week.

Days	Sun	Mon	Tue	Wed	Thu	Fri	Sat
No.of Accidents	14	16	8	12	11	9	14

- iii) The table below gives the number of aircraft accidents that occurred during the various days of the week. Test whether the accidents are uniformly distributed over the week.

Days	Mon	Tue	Wed	Thu	Fri	Sat
No.of Accidents	14	18	12	11	15	14

Link: <https://youtu.be/ynLXm4Gruu4?si=P0-bJKhtKwHf3WfS>

Type 4: Goodness of fit - Uniform Distribution:

The following is the distribution of the hourly number of trucks arriving at a company's warehouses.

Trucks arriving per hour	0	1	2	3	4	5	6	7	8
Frequency	52	151	130	102	45	12	5	1	2

Find the mean of the distribution and using its mean(rounded to one decimal) as the parameter λ fit a poisson distribution. Test for goodness of fit at the level of significance $\alpha = 0.05$

Link: <https://youtu.be/T3wpMYZj-ow?si=BqbEeIvBJi1ILcKM>

Type 5: Independence of attribution:

- 1) On the basis of information noted below, findout whether the new treatment is comparatively superior to the conventional one.

	Favourable	Non-Favourable	Total
Conventional	40	70	90
New	60	30	110
Total	100	100	N = 200

- ii) 1000 Students at college level were graded according to their I.Q and their economic conditions. What conclusion can you draw from the following data:

Economic Conditions	I.Q. Level	
	High	Low
Rich	460	140
Poor	240	160

Link: <https://youtu.be/yAfg5gRXVIU?si=Yw0RlXALZk1Vl-Zo>

- 2) Two researchers A & B adopted different techniques while rating the students level. Can you say that techniques adopted by them are significant?

Researchers	Below Average	Average	Above Average	Genius	Total
A	40	33	25	2	100
B	86	60	44	10	200

Total	126	93	69	12	300
--------------	-----	----	----	----	-----

Link: https://youtu.be/bAZ9za2g5_4?si=SS5156k2yf5wz-Ta

Type 6: Theory Questions

1. A sample analysis of examination results of 500 students was made. It was found that 220 students have failed, 170 have secured a third class, 90 have secured a second class and the rest, a first class. So do these figures support the general belief that the above categories are in the ratio 4:3:2:1 respectively?

II) A company carries out regular market research to obtain information about the customers. It is particularly important to know the age distribution of the customers. In the recent years this has remained fairly constant and is shown in the following table.

Age(Years)	% of Customers
Under 20	35
20-30	25
30-40	15
40-50	15
Over 50	10

A random sample of 300 customers selected during the latest market research investigation included the following numbers in each age group. Employ a Chi square test to examine whether there is an evidence of significant change in the age distribution.

Link: <https://youtu.be/x8M9mLSBCYw?si=XkQKcd3MX3IKBtGW>

Unit V – CORRELATION AND REGRESSION

Two Dimensional Random Variables – Correlation Coefficient:

- 1) Find Correlation coefficient from Table values:

x	65	66	67	67	68	69	70	72
y	67	68	65	68	72	72	69	71

Link : <https://youtu.be/w3cBih4LJaI?si=pwoSqcU2Yh1djRB9>

- 2) The two lines of regressions are $8x - 10y + 66 = 0$, $40x - 18y - 214 = 0$

The variance of X is 9. Find i) The Mean values of X and Y

ii) Correlation between X and Y.

- 3) Two random variables X, Y have the following joint probability density function is given by

$$f(x, y) = \begin{cases} 2 - x - y, & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Find the coefficient of correlation between X and Y.

Link: https://youtu.be/x1SIIfBpVXg?si=Kqr7D_ckjoE3O4K

- 4) The Joint Probability density function of two random variables X, Y is given by $f(x, y) =$

$$\begin{cases} \frac{6-x-y}{8}, & 0 < x < 2, 2 < y < 4 \\ 0, & \text{otherwise} \end{cases}$$

Find the coefficient of correlation between X and Y.

Link: <https://youtu.be/1IdHsI2ZGA4?si=jOQGJzb4GiIo3VoL>

- 5) If X and Y are two random variables having the joint probability density function $f(x, y) = 2$, for $0 < y < x$, $0 < x < 1$, find
- i) Marginal density function of X and Y .
 - ii) Conditional densities $f(x/y)$ and $f(y/x)$.
 - iii) Conditional variance of X given $Y = 2$,
 - iv) Check whether X and Y are independent or not?
 - v) Find $Cov(x, y)$
 - vi) Correlation between X and Y .

Link: <https://youtu.be/0u4QCZPbGsc?si=isf85G0cbIDPxYFu>

- 6) 2 Mark Questions Unit v

Link: [https://youtu.be/9wROzR0gTA4?si=yKfuxI25jkCLA -w](https://youtu.be/9wROzR0gTA4?si=yKfuxI25jkCLA-w)